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1. Assuming a simple 2-bit branch predictor that uses 8 bits from the PC to index into the BHT, how many entries does each BHT have? What is the size of the BHT (in bits)?

Using 8 bits to index means there are 2^8 entries. Each entry is 2 bit so the size of the BHT is 2^9 bit.

2. Assuming a (4,2) predictor, how many total entries does this predictor have? What is the total size of this correlating predictor (in bits)?

4 bits of global history, 2-bit counter, address not specified, so assuming 8.

Indexing size = 12
So BHT size = (2^12 entries) * (2 bit per entry) = 2^13 bits

3. What is the misprediction rate for the given traces and predictor?

	(0,1)	**(0,2)**	**(4,1)**	**(4,2)**	**(6,1)**	**(6,2)**
gcc-10K	30.58%	28.74%	20.12%	23.89%	20.29%	26.15%
gcc-8M	31.23%	28.17%	13.28%	11.15%	8.59%	7.31%

4. Is every entry in our branch predictor utilized? For our (4,1) predictor, how many entries are utilized? In order to answer this question, you will have to instrument your simulator to keep track of how many entries are used. Please update the following print statement in the printStats() function in predictor.cpp to print out the result

	(0,1)	**(0,2)**	**(4,1)**	**(4,2)**	**(6,1)**	**(6,2)**
gcc-10K	99.61%	99.61%	39.99%	39.99%	14.94%	14.94%
gcc-8M	100%	100%	83.67%	83.67%	48.62%	48.62%

Depending on 1) the length of the program 2) the number of table entries, the table shows different levels of utilization.

With longer programs, there is simply more history to keep track of, and the utilization goes up. With longer tables, branches have more unique indices, and not all indices are used, so utilization goes down.

5. How does the global branch history help improve branch prediction rates? Will all applications benefit from using global branch history?

Global history-based branch prediction is based on the empirical observations that branch outcomes are correlated because they often hinge on the same variables or even second or third-order relations between variables. The outcome of a branch using variable A can predict the outcome of other branches that use variable A as well.

If an application has branch outcomes that are mostly independent or noisy, global history can hurt accuracy.

6. What is a local predictor? How does it help with branch prediction?

A local predictor keeps per-branch history (a short shift register per PC) and uses that to index its table of counters. It helps when a branch's behavior depends on its own recent pattern (e.g., loops).

7. Explain what you would need to do to implement a tournament predictor in branchsim. The description should be concrete but concise (in 3-4 sentences).

A tournament predictor combines a global predictor and a local predictor in parallel, then uses a chooser table (2-bit counters that track which predictor was more reliable) to decide which predictor to trust for each branch history/PC. Afterwards, we update all three tables (global, local, and chooser) for each branch using the observed outcome.